

Introduction to Parametric Methods & Maximum Likelihood Estimation

Understanding model structures, parameters, and estimating truth from observed data.



Prescribed Textbook: Ethem Alpaydin, *Introduction to Machine Learning* (2004, PHI)



Teacher Note: Welcome students to Unit III. Today we transition from general ML basics to learning mathematical model parameters directly from training data.

Unit III Syllabus Roadmap

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Parametric Methods Intro
• TODAY'S FOCUS

2

Maximum Likelihood Estimation
• TODAY'S FOCUS

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Bias and Variance

4

Bayes' Estimator

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Parametric Classification

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Parametric Regression



What is a Parameter?

$$y = m \cdot x + c$$

m Slope parameter: controls tilt

c Intercept parameter: controls height

Definition

A parameter is an internal variable whose value controls model behavior.

Real-World Analogy

Knobs on an audio amplifier: adjusting bass and treble alters sound without changing amplifier design.



What is a Model?

“A model is a mathematical representation of a real-world relationship between variables in data.”

INPUT (X)
House Size (sq.ft)



MODEL (FUNCTION)
 $\text{Price} = m \cdot \text{Size} + c$

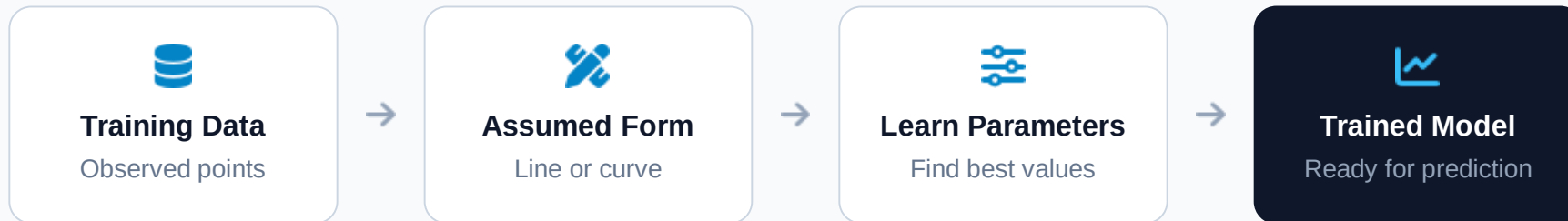


PREDICTION (Y)
Estimated Price (\$)



What is a Parametric Method?

“A parametric method assumes a particular model form and represents it using a fixed set of parameters.”



Machine Learning Form: $y = wx + b$

$$y = w \cdot x + b$$

x : Input feature

y : Target output

w : Weight parameter

b : Bias parameter

House Price Prediction

$$\text{Price} = w \cdot (\text{Size}) + b$$

Once w and b are fixed, any house size immediately gives an estimated price.

What Training Means

Training the model means finding optimal numerical values for w and b .



Why Parameter Estimation?

The Problem

We write down $y = wx + b$, but nobody tells us the values of w and b .

$w = ?$ $b = ?$

The Solution: Estimation

Use historical training samples to estimate the most sensible parameters.

Training Data → Parameter Estimation → Model

 **Key Method:** Maximum Likelihood Estimation (MLE) is the principal method used across machine learning to estimate parameters.

What is Estimation?

“Estimation is the process of guessing an unknown population parameter using information observed in sample data.”

OBSERVED SAMPLE (DATA)

H H T H H T H H

8 tosses: 6 Heads, 2 Tails

TARGET PARAMETER

$$P(\text{Head}) = \theta = ?$$

We cannot examine infinite tosses, so we use the 8 observed tosses to estimate the true probability.



Parameter vs. Estimator vs. Estimate

Term	What It Means	Coin Toss Example
Parameter (θ)	The unknown true property of the world	True bias $P(\text{Head})$ of the coin
Estimator ($g(D)$)	The mathematical formula / function used	Total Heads $\div$ Total Tosses
Estimate ($\hat{\theta}$)	The actual numerical value computed from data	0.75



What is Likelihood?

We observed 5 tosses of a coin:

H

H

H

T

H

(4 Heads, 1 Tail)

Hypothesis A

$$P(H) = 0.3$$

Very unlikely to yield 4 Heads

Hypothesis B

$$P(H) = 0.5$$

Moderate likelihood

Hypothesis C

$$P(H) = 0.8$$

Highest likelihood!



Probability vs. Likelihood

Probability

Parameter is **FIXED & Known**

“Knowing the coin has $P(H) = 0.5$, what is the chance of getting 4 Heads in a row?”

Evaluates: **Possible Future Data**

Likelihood

Data is **FIXED & Observed**

“Having observed 4 Heads in a row, which value of $P(H)$ best explains this outcome?”

Evaluates: **Candidate Parameters**



Maximum Likelihood Estimation (MLE)

“Maximum Likelihood Estimation chooses the parameter value that makes the observed training data most likely to have occurred.”

MAXIMUM

Find the peak or highest value among all possibilities.

LIKELIHOOD

How well the parameter explains observed samples.

ESTIMATION

Determining the unknown internal parameter value.



The 5-Step Process of MLE

- 1 Collect Observed Data (D):** Gather sample points (e.g., coin tosses, sales).
- 2 Define Likelihood Function $L(\theta | D)$:** Express data probability as a function of parameter θ .
- 3 Evaluate Candidate Parameters:** See which θ values generate high or low likelihood.
- 4 Maximize the Function:** Find the specific θ that gives the maximum value.
- 5 Output Estimated Parameter ($\hat{\theta}$):** Adopt $\hat{\theta}$ as the trained model parameter.



Coin-Toss MLE Calculation

OBSERVED EXPERIMENT

Tossed a coin **N = 10** times:

HEADS

7

TAILS

3

We want to estimate unknown parameter **p = P(Head)**.

MLE FORMULA APPLICATION

$$\hat{p} = (\text{Number of Heads}) \div (\text{Total Tosses})$$

$$\hat{p} = 7 \div 10 = 0.7$$

Conclusion: The MLE estimate for P(Head) is **0.7**. It represents the exact parameter under which 7 Heads and 3 Tails is most probable.



The Basic MLE Formula

$$\hat{\theta} = \arg \max_{\theta} L(\theta \mid D)$$

$\hat{\theta}$ Estimated parameter (the final winner)

$\arg \max$ Argument of maximum: parameter giving peak

$L(\theta \mid D)$ Likelihood: probability of data D given θ

D Observed training data sample



Why MLE is Fundamental in Machine Learning



Estimates distribution means and variances



Calculates class prior probabilities



Provides mathematical foundation for regression

The Parametric Learning Pipeline

1. Assume a parametric distribution
2. Write likelihood given training sample
3. Apply MLE to solve for unknown θ
4. Plug θ back into model to classify or predict



Quick Revision: 5 Key Concepts

PARAMETRIC METHOD Assumes a fixed mathematical model form before training.

PARAMETER Internal numerical variable defining the specific model curve.

ESTIMATION Calculating unknown parameter values from sample data.

LIKELIHOOD How well a candidate parameter explains the observed data.

MLE Selects the exact parameter value providing maximum likelihood.



Important Exam Questions

Short Answer Questions (2–3 Marks)

- Define a parametric method.
- Differentiate between an estimator and an estimate.
- Distinguish probability and likelihood.
- Define Maximum Likelihood Estimation.

Long Answer / Numerical (5–8 Marks)

- Explain the working steps of MLE with an example.
- Explain the basic formula: $\hat{\theta} = \arg \max L(\theta|D)$.
- Calculate MLE for a biased coin given 7 Heads in 10 tosses.
- Why is parameter estimation needed in ML models?

